

# Note 11.2 Calculus with Parametric Curves

## Tangents and Areas

- **Parametric Formula for  $dy/dx$**

A parametrized curve  $x = f(t), y = g(t)$  is **differentiable** at  $t$  if  $f, g$  are differentiable at  $t$ . If all three derivatives exist and  $\frac{dx}{dt} \neq 0$ , then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Here, we can understand the function  $y = f(t)$  as the composition of the functions  $y = h(x)$  and  $x = h(t)$ . Then we take derivative and we have  $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$  and it is the same as the upper form.

- **Parametric Formula for  $d^2y/dx^2$**

Since  $\frac{d^2y}{dx^2} = \frac{d}{dx}(y')$ , if the equation  $x = f(t), y = g(t)$  define  $y$  as a twice-differentiable function of  $x$ , then at any point where  $dx/dt \neq 0$  and  $y' = dy/dx$ ,

$$\frac{d^2y}{dx^2} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$$

## Length of a Parametrically Defined Curve

- **Definition of Length of a Parametrically Defined Curve**

If a curve  $C$  is defined parametrically by  $x = f(t), y = g(t), a \leq t \leq b$ , where  $f'$  and  $g'$  are continuous and not simultaneously zero on  $[a, b]$ , and  $C$  is traversed exactly once as  $t$  increases from  $t = a$  to  $t = b$ , then the length of  $C$  is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$$

- **Proof**

We subdivide the path  $AB$  into  $n$  pieces at points  $A = P_0, P_1, \dots, P_n = B$ . These points corresponding to a partition of the interval  $[a, b]$  by  $a = t_0 < t_1 < \dots < t_n = b$ , where  $P_k = (f(t_k), g(t_k))$ . Then the length of the line segment is

$$\begin{aligned} L_k &= \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \\ &= \sqrt{[f(t_k) - f(t_{k-1})]^2 + [g(t_k) - g(t_{k-1})]^2} \end{aligned}$$

Then by the Mean Value Theorem, there exists  $c_k, d_k$  in  $[t_{k-1}, t_k]$  such that

$$\begin{aligned}\Delta x_k &= f'(c_k)\Delta t_k \\ \Delta y_k &= f'(d_k)\Delta t_k\end{aligned}$$

Thus

$$\sum_{k=1}^n L_k = \sum_{k=1}^n \sqrt{[f'(c_k)]^2 + [g'(d_k)]^2} \Delta t_k$$

When the norm to zero, the  $c_k$  and  $d_k$  to the same. Thus

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \sqrt{[f'(c_k)]^2 + [g'(d_k)]^2} \Delta t_k = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$$

## Length of a Curve $y = f(x)$

For the curve  $y = f(x)$ ,  $a \leq t \leq b$ , we can let  $x = t, y = f(t)$ ,  $a \leq t \leq b$ . Then the length is

$$\int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{1 + [f'(t)]^2} dt$$

and it is consistent with the equations in [section 6.3](#).

## The Arc Length Differential

The differential of arc length is

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \sqrt{dx^2 + dy^2}$$

And the length is

$$L = \int ds$$

## Areas of Surfaces of Revolution

### • Area of Surface of Revolution for Parametrized Curves

If a smooth curve  $x = f(t), y = g(t)$ ,  $a \leq t \leq b$ , is traversed exactly once as  $t$  increases from  $a$  to  $b$ , then the areas of the surfaces generated by revolving the curve about the coordinate axes are as follows.

- For  $x$ -axis ( $y \geq 0$ )

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- For  $y$ -axis ( $x \geq 0$ )

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

## Exercises Section 11.2

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